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Progress in anti-biofouling materials and coatings for the marine environment

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Abstract: Marine biological fouling typically refers to the phenomenon of bacteria, algae, barnacles and other organisms attached to marine facilities, growing and reproducing. This occurrence can cause significant damage to ships, seawater nets, and other facilities, leading to economic losses and ecosystem damage. Existing biological fouling control technologies mainly include physical, chemical, and biological aspects, among which anti-fouling coatings are the most mainstream anti-fouling method in chemical control technology. This paper specifically analyzes the causes of marine biological fouling and the factors that influence it, and evaluates the advantages and disadvantages of the traditional biological fouling control technology from multiple perspectives, and discusses the future development direction of the new biological fouling control technology. At the same time, this paper provides an overview of marine anti-fouling technology from the perspective of the latest classification. And by summarizing and generalizing the cutting-edge anti-fouling coatings, it aims to provide inspiration for the application and development of biofouling technology in harsh marine environments.

Key Words:

Marine environment

Biological fouling

Control technology

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Introduction

Marine biofouling refers to the accumulation and attachment of marine organisms on marine structures and their adverse effects. This issue not only accelerates corrosion of materials (Zhang et al., 2016a), but also leads to problems such as increased resistance to ship travel and blockage of underwater pipelines (Lin et al., 2022; Su et al., 2020). These issues, in turn, interfere with the normal operation of the facilities, leading to significant economic losses and negative impacts on the natural ecosystem. In severe marine environments, like tropical regions with high temperature, humidity, and salt spray, monitoring instruments, ships, and oil platforms are more affected by fouling organisms. And materials submerged in seawater are especially prone to microbial attachment (Dineshram et al., 2009; Queiroz et al., 2024; Avelino-Jiménez et al., 2023).

Conventional antifouling coatings are limited by their toxicity to the marine ecology and environment. The trend in marine antifouling technology is moving towards non-toxic, functional, and sustainable solutions, driven by regulatory guidelines from the United Nations and others (Gao et al., 2022; Qiu et al., 2022; Wang et al., 2024). Existing antifouling control technologies necessitate innovative concepts, materials, and technologies. This requires interdisciplinary research that spans biology, physics, chemistry, materials science, and environmental studies. In addition, in the practical application of antifouling strategies, factors such as production efficiency, cost control, and compliance with environmental regulations must be taken into account. This process is challenging and requires continuous effort. Given the diverse and complex nature of severe marine biofouling, it is difficult to address all types of fouling with a single technology. Technologies such as modified superhydrophobic

interfacial materials, biomimetic materials and nanomaterials have emerged (Scurria et al., 2022; Ouyang et al., 2018; Harilal et al., 2020).

Although there have been many papers that systematically summarize marine antifouling technology, there is still a lack of overview articles on the latest antifouling coatings. Based on the latest discussions and the concerns of international environmental organizations, antifouling coatings can be classified into five main categories based on the presence or absence of biocides and the mechanism of antifouling action. The article will provide an overview of the principles, research progress, and applications of these five categories of coatings, while focusing on longer-lasting and greener antifouling technologies. Based on the characteristics and influencing factors of biofouling, this paper discusses the hazards of biofouling, and analyzes the influence of biofouling in severe marine environment by taking the South China Sea as an example. At the same time, the advantages and disadvantages of the traditional biofouling control technology and the development trend of the new biofouling control technology are elaborated. It also explores and forecasts biofouling control technologies in severe marine environments (**Fig. 1**).

2 Characteristics of biofouling

2.1 Definition and classification of biofouling

Marine biofouling is a phenomenon characterized by the undesired aggregation of marine organisms such as microorganisms, animals and plants and their growth on the surface of underwater facilities. It can be divided into two main categories based on the type of organisms involved and their impact on these facilities: microfouling and macrofouling (Duan et al., 2022). Microfouling organisms are subdivided into microbial film communities and microbiome communities, which primarily consist of microorganisms, bacteria, and protist. In contrast, macrofouling is dominated by mussels, seaweeds, barnacles, and other macroorganisms.

As depicted in **Fig. 2**, the formation of marine biofouling can be broadly divided into four stages:

Firstly, within a brief period (seconds to minutes), organic molecules such as

polysaccharides and proteins from seawater, facilitated by physical adsorption mechanisms and electrostatic interactions. They rapidly accumulate at the material interface, aiding in the formation of the initial conditioning film. After a few hours, microorganisms such as bacteria and unicellular algae, carried by seawater, begin to accumulate atop the pre-existing conditioning film, forming a more complex biofilm. Over several days and weeks, the microbial population release chemical signals that attract additional microorganisms, algal spores, and protozoan larvae. These organisms persist and flourish in this sticky coating, ultimately establishing microfouling communities. On the basis of the existing biofilm and microfouling communities, many larvae of macrofouling organisms in the planktonic stage select these microfouling communities as sites for forming a macroscopic layer dominated by calcified shell organisms.

However, the evolution of biofouling communities in real environments does not strictly adhere to the pattern outlined above. The diversity of marine organisms and the intricacy of ecosystems can influence the formation process, introducing variability into the sequence of attachment (Dang and Charles, 2015). Moreover, some large fouling organisms possess the ability to adhere directly to material surfaces not yet covered by biofilms, thereby bypassing the initial stages of the biofouling process.

2.2 Problems and hazards caused by biofouling

The physiological activities of fouling organisms, such as attachment, reproduction, and metabolism on the surface of marine engineering facilities, cause irreversible damage and lead to significant economic losses in the industrial sector. These organisms can directly interact with metals, dissolving them into ions, and together, these actions can initiate a destructive cycle of corrosion, biofouling, accelerated corrosion, and exacerbated biofouling (Al-Sultani et al., 2021; Jia et al., 2019; Batmanghelicha et al., 2017). This cycle is predominantly observed in carbon steel marine engineering structures and facilities, leading to phenomena such as algal blooms (Dong et al., 2022).

Specifically, the ship hull is a critical research area of study for marine biofouling, with

severe fouling conditions occurring on the propeller there are serious fouling conditions in the propeller (Farkas et al., 2021), below the waterline, and other hull sections. Beyond material corrosion, biofouling can also increase frictional resistance. Zou et al. (2023) demonstrated through simulation that under heavily fouled conditions, the additional resistance attributable to biofouling density, shape, and hardness can account for up to 80% of the total resistance. This increase in resistance results in higher carbon dioxide emissions and a degradation of environmental quality. Furthermore, fouling organisms can expedite the rate and progression of electrochemical erosion, intensifying metal dissolution and crevice corrosion by affecting the pH and dissolved oxygen in the attachment zone (Xu et al., 2023b; Liu et al., 2024c).

Secondly, marine fouling organisms prefer to live in groups, and mixed microbial communities are often more resistant than monoculture microorganisms (Ru et al., 2019; Zhang et al., 2019b). When they adhere to underwater facility piping systems, these organisms can reduce the flow rate of conveyance pipelines and decrease the efficiency of water delivery. In severe cases, complete blockage can lead to equipment failure (Li, 2023). When mariculture nets are exposed to inappropriate attachment of various types of marine organisms, they can significantly impede the nets' normal water exchange performance (Wu et al., 2021). Repairing and replacing damaged nets undoubtedly requires a substantial investment, in addition to the losses caused by production interruptions. These combined factors make biofouling a significant economic burden that restricts the sustainable development of the mariculture industry.

Additionally, marine biofouling can cause malfunctioning of marine observation facilities (Delgado et al., 2023). Unlike to the scouring effect that occurs when a ship is traveling, marine instrumentation equipment such as marine sensors, underwater acoustic instruments, and underwater cables, when at rest, pose significant safety risks (Salgar-Chaparro et al., 2020). Currently, Kong et al. (2017) have summarized maintenance methods for different types of equipment based on the signal principle and the impact of biofouling, among other factors, but the frequency of maintenance for equipment in deep and distant seas is still limited.

2.3 Characterization and challenges of biofouling in marine environment

It has been found that the temperature and salinity of seawater, the physicochemical properties of the materials on which organisms attach, and the dynamics of water currents all impact the attachment and growth of fouling organisms. Analyzing the growth of fouling organisms based on the specific conditions of different areas and controlling their growth through scientific means. Moreover, the formation process of biofouling is dynamic and ongoing, with each stage providing increased opportunities for subsequent organisms to attach and survive. Despite having gathered substantial information about surface-associated microorganisms, significant gaps remain in our understanding (Dang and Charles, 2015). Particularly, the processes and mechanisms of surface colonization and biofilm development by these microorganisms are highly complex, involving community structure, function, and their impact on the dynamic marine environment. Fouling communities can develop rapidly, necessitating efficient and enduring antifouling measures.

It is important to note that addressing biofouling is more critical than rebuilding marine structures. Traditional toxic anti-fouling coatings, which effective, are widely used and may pollute marine ecosystems, affecting non-target organisms. Although new antifouling materials are emerging, their stability and economic viability require further validation.

3 Effects of biofouling in severe marine environment

3.1 Factors affecting biological fouling

Marine environmental factors are crucial to the occurrence of biofouling, with temperature being the most critical determinant of its distribution. The differences in fouling communities essentially reflect temperature characteristics (Lin et al., 2017). The species, attachment stage, and rate of fouling organisms are all intimately linked to water temperature. Based on the degree of temperature adaptation, fouling organisms are categorized into warm-water and cold-water species, with the former predominating.

In addition to temperature, the attachment of fouling organisms is influenced by a combination of local natural environmental conditions, such as pH, salinity, hydrodynamics,

and dissolved oxygen, as well as human activities and other factors (Marciales et al., 2019; Zhang et al., 2019c; Portas et al., 2023; Zhang and Wu, 2020). Please refer to **Appendix A Text S1** for detailed information. Salinity can directly affect the osmotic pressure within fouling organism cells. Furthermore, variations in aquatic flow rates significantly impact the growth and attachment of fouling organisms, particularly algae. Changes in water flow can alter the concentration of various substances. Additionally, metabolites produced by organisms during their metabolic processes may inhibit their growth; a well-regulated water flow can help disperse these residues, thereby increasing the organisms' growth rate. At the same time, the adsorption behavior of fouling organisms is influenced by various characteristics of the material surface, such as color, roughness, and tilt angle in the marine environment.

3.2 Impacts of biological fouling in severe marine environment

Taking the South China Sea area as an example, it is considered a typical severe marine environment due to its high temperature, high humidity, high salt and strong radiation. The biofouling problem of marine equipment and infrastructure in service is prominent, which seriously affects the normal use. This is a key area for fouling biology research in China (Ding et al., 2020). Due to being located in the low-latitude tropical region, with year-round high temperatures, the fouling organism community has high diversity and rapid reproduction rate (Ma et al., 2024), featuring a multitude of attached organisms and extensive coverage areas. In addition, the South China Sea, excluding a few estuarine waters, has very high oxygen content and is highly saline, and predominantly supports fouling organisms that tolerate high salinity. The fouling organisms in the South China Sea mainly include trachypods, mollusks, mosses and macroalgae, among which, barnacles and mosses are the two most common fouling organisms in the current study (Wu and Ding, 2023; Chen et al., 2023). Studies by Peng et al. (2019) and Wang et al. (2021) on aluminum alloy and copper alloy materials located in the Sanya Sea, found that the area covered by large marine organisms is far more than that of Qingdao and Zhoushan areas, where surfaces are almost entirely covered by fouling organisms, and serious crevice corrosion due to acidification of medium caused by decomposition of

organic matter in barnacle shells. At the same time, biofouling and material corrosion are inseparable and they interact with each other (Zhang, 2017). Biological attachment sites can form an oxygen concentration difference with unattached sites, accelerating metal corrosion at the attachment sites, and further promoting metal dissolution through pH reduction in localized occlusion zones (Guo et al., 2022). As a result, pitting corrosion is also more severe in areas with strong growth of marine organisms and high attachment rates.

Tropical environmental problems also plague the Indian Ocean, the Caribbean Sea and other seas. Again, these areas are severely jeopardized by biofouling because of their high growth rates. It has been suggested that physical removal of biofouling should be avoided in the nearshore Caribbean Sea. Due to the unique climatic characteristics of tropical seas, the fouling organisms left behind are able to rapidly proliferate. Residual macro-organisms can reproduce rapidly as invasive species, destroying ecosystems. Many non-native corals have already appeared in the Caribbean Sea as a result (Hoeksema et al., 2023). Therefore, it is extremely important to carefully analyze the temperature and salt characteristics, hydrodynamic features, physicochemical properties of the attached surface materials, and the composition of fouling organisms of the severe marine environments, which is very crucial for improving the biological fouling control technology.

4 Biological fouling control techniques

4.1 Advantages and disadvantages of traditional biological fouling control methods

Traditionally, various techniques have been employed to keep metallic materials free from fouling organisms. These include mechanical removal, the application of antifouling coatings, and the incorporation of alloying elements (He et al., 2023a; Cai et al., 2020; Ding et al., 2019). The most prevalent practice has been the effective management through antifouling coatings containing metal compounds or biocides (Zhang et al., 2023; Muthukrishnan et al., 2017). These coatings can rapidly and immediately reduce bio-attachment, yielding significant short-term results. Over the years, these technologies have been studied extensively and are now mature, smooth, and easy to apply in the industry.

Despite the variety of anti-fouling strategies in use, they all aim to interfere with or block the adhesion and reproduction mechanisms of fouling organisms. The goal is to fundamentally weaken their ability to adhere or inhibit their growth and reproduction processes, thereby minimizing the adhesion of fouling organisms and preventing further development. It is crucial to note that the initial attachment of biofilms forms the basic framework for the attachment of subsequent macro-organisms, such as barnacles and mussels (Yu et al., 2022). Biofilms act as a 'bridge', releasing specific chemical signals that guide marine organisms in assessing environmental suitability and influencing their survival decisions. Thus, the key to antifouling is early intervention, controlling biofouling at its early stages to curb the adhesion and growth of marine organisms (Zhang and Wu, 2020).

However, these traditional strategies come with several drawbacks, including high toxicity, high costs, environmental pollution, poor dispersion, and difficulty in control (Wu et al., 2023a; Adeleye et al., 2016). They have significant negative impacts on the ecosystem and human health, limiting the potential for effective biofouling control. The long-term release of heavy metals or toxic chemicals in some conventional antifouling paints can lead to water pollution and ecosystem damage. Moreover, the biocidal substances are not targeted; they may harm non-target organisms, causing significant impacts on aquaculture crops. Over time, these chemicals are persistent and bioaccumulative, posing a potential threat to humans through long-term accumulation in the food chain.

4.2 Trends in new biological fouling control technologies

In recent years, with the gradual increase in environmental awareness, the development of green antifouling coatings with sustainability, broad-spectrum antifouling properties, and non-toxicity has become imminent. The development trend of novel biofouling control technologies focuses on improving efficiency, reducing environmental pollution, enhancing sustainability, and realizing longer-term protective effects. The main directions regarding their development include the following four aspects (Zhao, 2023; Liu et al., 2022; Almeida et al., 2015).

4.2.1 Environmentally Friendly Materials

Fouling organisms typically adhere tightly to the surface of a target substrate or to an existing layer of fouling through chemical binding, electrostatic interactions, and physical adsorption. Theoretically, the use of low-surface-energy coating technologies can effectively impede the attachment of fouling organisms to material surfaces. The molecules (atoms) in the surface layer of the solid phase are in a spatially asymmetric force field, the liquid phase molecules exert less force on them, and the molecules (atoms) are subjected to tensile forces directed towards the interior of the solid phase. Since the molecules of the surface layer are subjected to inward pulling forces and the expansion of the surface area requires the environment to do work on the system, an increasing surface area of the system necessarily increases the energy. According to the energy minimum principle, this system is unstable, so the surface area of the system will take the smallest value possible. Low surface energy coatings reduce the spreading and adhesion of liquids and solids on their surface by lowering the surface energy. The lower the surface energy, the higher the contact angle of the liquid on the surface and the lower the adhesion. In summary, low surface energy coatings make it difficult for the mucus secreted by marine organisms to wet the surface of the coating, which is not conducive to the attachment of marine organisms to the surface of the coating.

Even if initial attachment occurs, the bonding strength is so weak that it can be dislodged under slight hydrodynamic shear stress. This realizes the self-cleaning effect of the ship's surface layer and achieves the purpose of antifouling, embodying the concept of a desorption-based anti-fouling coating (Cox et al., 2022). However, lower surface energies are not always better. Baier has proposed a relationship between the surface energy of a material and the amount of biofouling by means of the “Baier curve”, which shows that surface energies between 22 and 24 mJ/m² minimize the amount of biofouling. Therefore, this type of process does not focus only on reducing surface energy. It should also be considered in the overall context of the Baier curve, environmental factors, and cost.

The optimization of the modification process of harmless low surface energy coatings emphasizes the combination of physical methods, ecological design, and biological control to

enhance their ability to resist the attachment of marine organisms and the material's strength (Zhai et al., 2023; Liu et al., 2020). The use of such materials can reduce reliance on harmful chemical antifouling agents. Degradable and recyclable materials ensure natural decomposition after the antifouling cycle, reducing waste. Self-repairing coating technologies are also being developed to extend service life.

4.2.2 Bio-inspired and Biomimetic Technologies

Over the past decades, traditional toxic antifouling coatings have been gradually banned globally. As an alternative, more and more bio-inspired biomimetic antifouling coatings are being manufactured and tested (Jin et al., 2022) (**Fig. 3**). Most biomimetic antifouling measures focus on superhydrophobic technology. Superhydrophobicity is controlled by surface roughness, free energy and chemical homogeneity.

Rough surfaces can trap air in the grooves and stabilize the superhydrophobicity through the Cassie-Baxter model. This air trapping will amplify the surface hydrophobicity and reduce the free energy of the surface. The Cassie-Baxter model is applicable when the surface is partially wetted and there is air trapped between the liquid and the surface (Selim et al., 2019, 2023b). Cassie-Baxter's formula is as follow:

$$\cos\theta_{CB} = f_s(\cos\theta_s + 1) \quad (1)$$

where, θ_{CB} represents the contact angle on surfaces with microstructures. θ_s is the contact angle on smooth surfaces. f_s is the ratio of the area of the solid surface in contact with the liquid to the total contact area. An increase in surface roughness leads to an increase in the contact angle and an increase in the actual surface area. This reduces bioattachment by decreasing the area of liquid attachment to the surface and increasing the difficulty of spreading. When the surface roughness of a low surface energy material is increased, the surface energy can be further reduced, which enhances fouling protection (Selim et al., 2016, 2022, 2023a).

In addition, the Wenzel model applies when the surface is completely wetted by the liquid. According to the Wenzel model, the relationship between the contact angle of a rough surface and that of a smooth surface is linear. Wenzel's formula is as follow:

$$\cos\theta_w = r\cos\theta \quad (2)$$

where, θ_w is the contact angle on a rough surface. θ is the contact angle on a smooth surface. r is the surface roughness factor, defined as the ratio of the actual surface area to the projected surface area. If the smooth surface is hydrophilic ($\theta < 90^\circ$), the rough surface further enhances hydrophilicity; if the smooth surface is hydrophobic ($\theta > 90^\circ$), the rough surface further enhances hydrophobicity.

Nanostructures covering the surface of the lotus leaf ensure superhydrophobicity and low adhesion. These microstructures form a physical barrier for the surface substrate to resist biofouling organisms and significantly reduce fluid resistance due to air retention effects that reduce boundary layer viscosity (Chen et al., 2021). Thus, microstructures can exhibit a dual effect if applied to ships. Actively exploring the development of similar superhydrophobic and biomimetic materials with superior anti-bioadhesion properties is crucial. Drawing on nature's anti-fouling mechanisms, such as mimicking the microstructure of dolphin's skin to create self-cleaning surfaces, or developing superhydrophobic materials similar to the lotus leaf effect, can reduce bio-attachment (Zhang et al., 2016b; Wu et al., 2023b).

However, superhydrophobic structures are prone to failure. The physical structure of its surface can be damaged by pressure. Low surface energy molecules are decomposed by external stimuli such as heat and light. In the presence of large fluctuations, the air layer is prone to escape. Therefore, there have been many studies on multilayer micro and nano structures to protect the air layer.

Environmental conditions vary significantly from sea to sea, and we need to develop and apply bionic antifouling technologies in a targeted manner. Offshore waters are characterized by high biomass, and antifouling coatings need to respond quickly. The application of superhydrophobic technology similar to shark skin and other structures is very important. At the same time, in order to prevent rapid fouling of marine facilities under static conditions, the addition of certain biocides is critical. Deep and distant sea equipment needs to focus on the application of long-lasting control technology. As a result, short-term superhydrophobic technologies remain unpopular. Due to temperature, salinity, pressure and other effects, the

application of bionic hydrogel is very limited in the sea area.

However, no single biomimetic antifouling method can be identified as a universal strategy in severe marine environments. Scholars need to address the limitations of single biomimetic strategies by developing multifunctional biomimetic antifouling strategies and combining the advantages of various biomimetic and non-biomimetic antifouling strategies to enhance performance and durability.

4.2.3 Smart responsive materials

Many scholars have developed smart materials that can adjust their surface properties in response to environmental changes, such as light, pH, temperature, or biological signals (Ren et al., 2023). Most coatings use light-responsive signals to develop inorganic photocatalytic materials based on materials like titanium dioxide and zinc oxide (Liu et al., 2024b). These materials produce strong oxidizing free radicals under light, which destroy the cellular structure of fouling organisms and achieve antifouling effects. Detailed information are provided in **Appendix A Text S2**. Additionally, Tong et al. (2022) prepared a smart marine antifouling polyurethane coating with UV(Ultraviolet)-responsive controlled release of coumarin, inspired by sea slug defense. This strategy reduces unnecessary chemical release while effectively resisting biofouling to achieve precise delivery of antifouling agents and efficient antifouling efficacy, especially for the specific antifouling needs of complex piping systems, particularly those made of titanium alloy.

In tropical areas, the application of photosensitive materials can enhance the anti-fouling effect in sunny conditions. In polar regions, the use of low-temperature resistant shape memory materials will achieve better results. In offshore waters, barnacles and mussels are prevented from attaching by pH-responsive materials.

4.2.4 Multifunctional integrated antifouling

Combining physical, chemical, and biological antifouling mechanisms into a single system, such as the integration of nanomorphous alloys with the slow release of biologically active

substances in dual-action antifouling coatings, offers a comprehensive approach. This includes synergistic sterilization with compound antifouling agents and external fields, as well as composite coatings and electrocatalytic hydrogen peroxide generation (Zhai et al., 2020; Wen et al., 2023; Wang et al., 2020). These methods provide more comprehensive protection. For instance, using corrosion-resistant stainless steel as the main body, with anti-fouling coatings serving as a barrier, can extend the life of the structure in a holistic manner (Singh et al., 2024). As application environments become more complex, multifunctional integrated fouling solutions are gaining popularity. They not only meet basic antifouling requirements but also incorporate features like abrasion resistance and corrosion resistance. Moreover, these solutions are more environmentally friendly, aiming to address the comprehensive challenges of fouling in specialized areas.

The trends outlined above indicate that future anti-fouling technology will concentrate on material innovation, intelligent design, and multifunctional integration. Please refer to **Appendix A Text S3** for detailed information. This focus aims to meet the profound needs of marine development from a more ecological perspective and to foster the sustainable, green advancement of marine equipment technology. Compared with traditional anti-fouling technology, new anti-fouling technology accomplishes performance upgrading by means of multifunctional combination and other means. Especially in terms of environmental protection, superhydrophobic technology, self-polishing functions and nanostructures play a key role. They can not only be applied on large structures, but also complete the protection of shaped structures.

Although the direction of development at home and abroad is generally consistent, there are differences in research priorities. Foreign countries are committed to the research and development of smart materials and smart devices. Although domestic scholars are also involved in these fields, the focus is still on coating modification. Committed to environmental protection, the domestic environmental protection coatings are developing rapidly. However, the market share of domestic coatings is still small. We should focus on the development of high-performance and comprehensive antifouling technology.

However, in developing environmentally friendly marine antifouling coatings, the

economic costs and environmental impact during the production process must be carefully weighed. High production costs can hinder commercialization, and the use of toxic chemicals along with the discharge of toxic by-products may threaten the environment. Even when employing natural antifouling agents, their environmental compatibility should be thoroughly assessed, despite their natural occurrence. Currently, many biomimetic antifouling coatings have only been tested under laboratory conditions and have not yet transitioned to real-world applications. In practical use, novel materials may face various limitations. In conclusion, while new marine antifouling coatings hold great potential, they also confront significant challenges and opportunities in the field of application.

4.3 Classification and scope of application of biofouling control technologies

Current biofouling management centers on limiting the attachment of fouling organisms, especially controlling the growth of clusters of large organisms on the surface of underwater facilities to mitigate their negative impacts. While some of the existing techniques are effective in targeting adult macrofouling organisms, most of them still focus on intervening in their larval or early life stages. Detailed information are provided in **Appendix A Text S4**. Given the diversity of marine fouling communities and complex ecosystems, no single technology has yet been able to comprehensively address the full range of fouling problems. At the operational level, the choice of anti-fouling strategies needs to balance a number of considerations, such as productivity, economic costs and compliance with environmental protection regulations. In reality, a variety of physical means, chemical biocides and anti-fouling coatings have been widely adopted to achieve the goal of prevention and control of biofouling (Duan et al., 2020).

4.3.1 Physical antifouling

The primary issue is the prevention or removal of biofouling through physical means. Current physical intervention strategies for facilities such as mariculture tanks, ship hull exteriors, and large-scale seawater recirculation cooling systems encompass mechanical removal, including the use of submersible robotics (Su et al., 2024). Other measures include seawater purification

and filtration, water flow rate regulation, temperature treatment processes, ultrasonic interventions, ultraviolet radiation for antifouling, osmotic pressure regulation strategies, and reduction of oxygen content.

The removal of hard-attached organisms like mussels and barnacles from hull shells and net coats typically involves scraping, combining human labor with mechanical equipment. Notably, the use of underwater robots has evolved from being diver-controlled to utilizing autonomous removal equipment. Concurrently, high-pressure water jet cleaning has become a popular cleaning technology. Physical antifouling techniques such as mechanical removal and ultrasonic waves, while effective, have some limitations. For example, mechanical removal may cause damage to the substrate. Ultrasonic and UV technologies have high energy consumption for removal. Fouling removal techniques for underwater robots require regular maintenance and replacement of components. In addition, large-scale labor costs account for a certain percentage. Moreover, all these types of removal techniques have certain safety risks. In terms of long term effectiveness, physical control technology does not have an advantage.

4.3.2 Chemical anti-fouling

Generally speaking, chemical antifouling refers to the use of specific chemical measures or agents to avoid, actively inhibit, or kill marine organisms and their spores, larvae. The goal is to prevent marine organisms from attaching to or gathering on the surfaces of underwater facilities. Chemical antifouling is mainly categorized based on the different means used, such as the chemical agent method, electrolytic antifouling method, and the application of antifouling paint.

The chemical method offers several advantages, including ease of application, high antifouling efficiency, and a broad range of applications. It is currently the commonly adopted method in industries such as shipping and coastal power plants. Current antifouling agents consist of oxidizing and non-oxidizing compounds. Oxidizing compounds damage important structures of microbial cells, while non-oxidizing compounds target specific cell components, such as cytoplasmic constituents. In addition to traditional chemical antifouling agents,

modified hydrogel coatings have gained popularity in recent years as a green marine antifouling solution. These coatings significantly slow down or even completely inhibit the growth of marine organisms on the substrate surface by effectively preventing the adsorption of proteins and polysaccharides (Liu et al., 2023b).

Chemically controlled techniques present the challenge of preparing component materials that are difficult to balance. Many of the prepared materials are typically limited to the gram scale, making it difficult to scale up from laboratory studies to applications. In addition, marine species are so diverse that biomimetic anti-marine fouling materials face significant challenges in terms of broad-spectrum fouling performance and long-lasting fouling protection. Some special materials show excellent anti-fouling performance, but the deep relationship between fouling and interface needs to be further explored. At the same time, certain properties may change in the natural marine environment. Therefore, there is a need to increase the self-responsiveness of materials to changes in the external environment. More importantly, the compounds are usually toxic. Modern research therefore prefers to reduce their biocide content or use non-toxic technologies. However, this means that their cost, longevity, and broad-spectrum properties need to be studied in depth.

4.3.3 Biological antifouling

Typically, this method harnesses the predatory or competitive relationships between organisms to suppress the growth and reproduction of fouling organisms, either directly or indirectly, by introducing specific species (Chen et al., 2023). Despite being environmentally friendly, its application is currently less common due to several limitations: the limited number of controllable fouling species, the lack of a mature mixotrophic mode, and the potential risk of ecosystem changes associated with the introduction of exotic species.

5 Advances in biofouling control technology in severe marine environments

5.1 Improvement of intrinsic tarnish resistance of metal substrates

Specially processed alloy fillers offer significant advantages in large-scale industrial

applications. The incorporation of certain rare earth elements can enhance the antimicrobial properties of these materials (Hao et al., 2023; Geng et al., 2024). Composites with high-temperature and corrosion-resistant properties (Zheng et al., 2024; Sun et al., 2024b), are utilized in complex and severe marine environments. In recent years, metal elements and metal-based compounds have been commonly used to fabricate composites through microstructural modification or alloying (Ahmadkhaniha et al., 2016). Please refer to **Appendix A Text S5** for detailed information. For instance, Ti-Cu alloys, once modified, exhibit excellent mechanical and antimicrobial properties (Arroussi et al., 2021; Javadhesari et al., 2020; Zhang et al., 2016b; Ma et al., 2022; Lu et al., 2021). On nickel advanced weathering steels exposed to seawater, product films with a bilayer structure have appeared. Despite some defects, these films have a potential difference between their inner and outer layers. The negatively charged inner product film can block the erosion caused by Cl^- in the salt spray environment (Wu et al., 2017). Magnesium alloys are known for their high specific strength, low density, biocompatibility, and electromagnetic shielding (Huang et al., 2023; Wu et al., 2024b). As a green engineering material for the new century, they have vast development and application potential across various industrial sectors, including automotive, marine, and aerospace (Liu et al., 2023a; Sun et al., 2024a; Jiang et al., 2022; Li et al., 2023b). Metal frameworks constructed by coordination with organic ligands have garnered widespread attention due to their flexible structure and adjustable functions. These frameworks ensure strong antimicrobial properties and hold the potential for antifouling applications in severe marine environments and industrial wastewater (Wen et al., 2019).

Research on industrial pure titanium at the Sanya Marine Environmental Test Station by Ma et al. (2018) revealed that the titanium oxide film generated is dense and stable, suitable for use as an important anticorrosion and antifouling material in the South China Sea area. However, it was observed that the effectiveness of this film is compromised when the fouling community or local pH changes, indicating that the issue of marine biofouling on titanium alloys requires urgent further study and resolution. Additionally, Guan et al. (2017) identified unique microbial community compositions for different types of alloys in highly corrosive steel in Sanya, China.

Therefore, the development of specific antifouling materials and a deeper understanding of their antifouling properties for particular marine environments are still needed. The corrosion tendency of these materials is also a critical factor to consider (Li et al., 2023a; Shi et al., 2023; Jiang et al., 2024).

However, metal matrix modification usually involves complex manufacturing processes, resulting in high production costs. In marine environments with high flow velocities and particle impacts, their wear resistance still needs to be further improved. Moreover, the anti-fouling effect of pure metal still has some defects.

5.2 Physical control technology

5.2.1 Ultrasonic technology

Ultrasonic technology, as a non-chemical method, performs biofouling control by emitting sound waves without introducing harmful substances. This technique leverages the propagation properties of ultrasound waves in water to create physical effects that disrupt the attachment process of organisms (**Fig. 4**). It minimizes the aggregation of fouling communities, such as algae and shellfish, on critical surfaces of structures, offering a new approach to ecologically friendly biofouling management strategies. Hotrum et al. (2015) summarized the use of ultrasound as a mechanical effect that enhances heat transfer, mass transfer, and membrane flux processes. The inertial cavitation that occurs during the process can be attributed to the extreme temperatures and pressures in its localized area, which can scour the surface and inactivate microorganisms. The findings of one study demonstrated that ultrasound can effectively control the biomass of immature shell stages of mussels (Zhou et al., 2020).

Ultrasonic control is commonly utilized for cleaning marine facilities, such as ship hulls. However, there is a risk of equipment surface corrosion due to continuous exposure to ultrasound, as well as potential negative effects on other protein-functional substances. Additionally, ultrasonic frequencies may fall within the hearing range of some marine organisms, potentially leading to adverse outcomes (Legg et al., 2015). Thus, there remains significant potential for further exploration into the practical applications of ultrasound

technology.

5.2.2 UV irradiation

UV irradiation is an established disinfection procedure that is chemical-free and has minimal by-products. It is effective against most bacteria, viruses, and protozoa, easy to perform, and cost-effective. UV irradiation resulted in less protein accumulation, a cleaner membrane surface, and increased porosity, which ultimately resulted in a thinner and less homogeneous biofouling layer. Numerous studies have shown that UV light can significantly affect the structure of bacterial communities on membrane surfaces (Feng et al., 2023; Braga et al., 2020b). This method is primarily applied to fully submerged areas below the waterline of marine structures and ships in harbors. For ships operating in severe marine environments, installing UV lamps on the hull for biofouling control has been a partial option. Pucherelli et al. (2018) have demonstrated that the maintenance costs for coolers affected by mussel and bacterial fouling can be reduced by 75 % under the action of a medium-pressure UV light system. However, low UV transmittance is a challenge for successful inactivation, and it is important to balance UV energy density while considering the drag effect on inactivation. In addition to low-pressure and medium-pressure vapor lamps, UV-LED emitters developed in recent years have significantly delayed biofilm formation, although they do not have significantly higher biofilm inactivation efficiencies (Sperle et al., 2023).

As mentioned above, UV irradiation has been shown to be effective in preventing biofouling, but there are still many unknowns regarding its use as a hull fouling prevention method. Specifically, the sufficient exposure time to control biofouling, the optimal distance of the UV irradiation source for biofouling control, and the impact of UV light on established fouling communities are areas that require further investigation. Therefore, the team of Cierra Braga has designed a device for application in future field studies to explore differences in fouling communities with respect to coating type, UV intensity, frequency, and exposure duration (Braga et al., 2020a).

5.3 Chemical control technology

According to the latest standards, marine antifouling coatings can be divided into the following five main categories.

5.3.1 Self-polishing antifouling coatings with biocides

Self-polishing coatings are typically based on copolymers that contain both hydrophilic and hydrophobic segments. The hydrophilic component is responsible for the erosion of the coating, while the hydrophobic segment ensures the controlled release of biocides. As the ship moves through water, the hydrophilic segments dissolve, exposing fresh layers of the coating and releasing biocides. The process is designed to be slow and consistent, ensuring a steady release of biocides over time. This kind of coating is primarily applied in the area where SPCs help reduce drag and fuel consumption. And it can be used on oil rigs, wind turbines, and other offshore installations to prevent fouling (**Fig. 5**).

Hu et al. (2024) prepared composites by integrating zinc acrylate resin with indole derivatives for controlled release of antifoulants. The coating inhibits the growth of algae in two aspects, firstly by inhibiting the quorum sensing and virulence factors in the fouling organisms through indole derivatives, and at the same time disrupting the balance of calcium ions in the algal cells. Secondly, the hydrophilic group of the zinc acrylate resin is continuously dissolved and new indole derivatives appear to achieve a self-polishing effect on the surface. This facilitates easy detachment of attached organisms.

Cuprous oxide is widely used as the most mainstream antifouling agent for self-polishing coatings. Zhang et al. (2022) investigated the specific properties of coatings respectively containing Cu_2O and CuSCN . It was found that the self-polishing coatings containing Cu_2O and CuSCN could achieve a stable protective performance for 4 years in the immersion experiment. However, the dissolution of the topcoat containing CuSCN produces relatively larger voids and thicker leaching layers, suggesting that the Cu_2O -containing coating system has better barrier properties. In addition to copper compounds, studies have been conducted on the application of plasma sprayed Cu/Ti hybrid powder filling self-polishing coatings. This Cu-

Ti composite coating achieves its antifouling properties by controlling the release of Cu ions through galvanic dissolution of the Cu laminates. This slow release and self-polishing effect resulted in a lifetime two times longer than conventional self-polishing coatings (Tian et al., 2021).

By now, advances in polymer chemistry and nanomaterials are driving the development of more effective and sustainable SPCs. Ongoing research focuses on minimizing the ecological footprint of antifouling coatings. Researchers are trying to evaluate the environmental and economic impacts of SPCs throughout their lifecycle to guide sustainable development. Self-polishing antifouling coatings with biocides are suitable for ships at different speeds and in different waters, including low-speed navigation modes and warm waters. These coatings usually have some mechanical strength. Although the release of biocides has been somewhat controlled, their longevity has yet to be studied in depth. The release of biocides may be explosive. These coatings are expensive to develop.

5.3.2 Non-self-polishing coatings with biocides

Despite the gradual rise of self-polishing coatings, there are still some non-self-polishing coatings that occupy a certain position. Due to its excellent antimicrobial properties and abrasion resistance, copper-based coating remains the most widely used contemporary solution for biofouling control. Copper oxide has become one of the most prevalent antifouling biocides in marine coatings, providing effective fouling control on the majority of steel hulls globally.

Recent studies have indicated that modern copper-based antifouling coatings are free from metallic copper, containing instead cuprous compounds and copper oxide. This allows for their safe application over epoxy barrier coatings on aluminum substrates (Chasse et al., 2020). Xu et al. (2023a) have demonstrated the effectiveness of copper-based coatings. Detailed information are provided in **Appendix A Text S6**. Due to their excellent antimicrobial properties and abrasion resistance, these coatings remain the most widely used contemporary solution for controlling biofouling.

Nanostructure coating technology involves incorporating nanoscale materials, with at least

one dimension limited to the 1-100 nm range, into conventional coating systems. It leverages the unique dimensional effects of these nanomaterials, such as small size effects, interfacial effects, quantum confinement effects, and dielectric-domain-limiting effects, to enhance the coatings' properties (He et al., 2023b). As shown in **Fig. 6**, research and applications have demonstrated that nanomaterials have been successfully applied in various biofouling control endeavors within severe marine environments. These coatings serve key functions, including antimicrobial, antifouling, waterproofing, and weather resistance (Zhang et al., 2019a; Selim et al., 2017; Ellinas et al., 2021; Jesus et al., 2021).

In recent years, the incorporation of various nanomaterials, control of particle content, and combination with superhydrophobic materials have enabled coatings to meet specific objectives. However, challenges remain, such as high costs, potential biotoxicity, reduced mechanical strength, and difficulty in removing long-standing biofouling (Toozaandehjani et al., 2024; Chen et al., 2022). Currently, the testing cycle for nanocomposite coatings is relatively short. Future research should not only address these issues but also investigate synergistic anti-fouling mechanisms to ensure the longevity and universality of nanostructure coatings.

Non-self-polishing coatings have diverse applications in the marine industry, including marine antifouling, drag reduction, corrosion prevention, oil-water separation, and marine sensor technologies. However, challenges such as achieving precise reproduction of microstructures, ensuring long-term stability, and addressing economic feasibility for widespread adoption remain.

5.3.3 Fouling self-peeling antifouling coatings with biocides

Fouling self-peeling coatings represent a significant advancement in the field of marine coatings, designed to minimize the adhesion of marine organisms to submerged surfaces. One of the principles of action of certain compounds used in self-release applications is the modulation of surface free energy as well as roughness. Although the relationship between low surface energy and roughness still needs to be studied in depth. However, it is certain that there is a close link between the antifouling properties of a coating surface and the roughness. Some

of the nanogroove structures can trap air in the voids, making the surface appear superhydrophobic. The antifouling principle of the above coatings is often referred to as the Cassie-Baxter model. Unlike traditional biocidal antifouling coatings, they rely on surface properties such as low surface energy, smoothness, and elasticity to prevent fouling. Fouling self-peeling coatings work by creating surfaces that are hard for fouling organisms to stick to. The primary mechanisms involve low surface energy, smoothness, and elasticity, aiming to achieve low adhesion strength to minimize the force needed to remove fouling organisms. These coatings also release biocides to completely eliminate biofouling from facility surfaces and prevent the attachment of fouling organisms. Although both fouling self-peeling coatings and fouling self-release coatings achieve fouling protection through low surface energy and flexible properties, self-peeling coatings place more emphasis on the mechanical properties of the coatings, their compatibility, and their environmental effectiveness. Progress in research on this type of coatings is described in detail in the following section.

Eugenol, as a powerful natural antifouling agent, is gradually being widely used as a substitute for metal ions. Liu et al. (2024a) synthesized a novel fluorinated eugenol copolymer and covalently grafted it onto modified PDMS to accomplish the controlled release of eugenol. The coating can achieve significant antifouling, anticorrosion and strong weather resistance in the marine environment. Chen et al. (2024) demonstrated that polybenzoxazines are an excellent material for self-peeling antifouling paints by changing the structure of the compounds. In addition to this, some scholars have grafted polyethylene glycol and quaternary ammonium salts into cross-linking structures to anchor them to the surface of the coating. This coating has tremendous antimicrobial efficiency and antidiatom performance (An et al., 2024) (**Fig. 7**).

However, the fouling self-peeling antifouling paint has poor antifouling ability under static conditions and is susceptible to mechanical damage, which is difficult to repair. And functional modification in it will reduce its mechanical properties and increase the cost, future research should tend to lift this limitation.

5.3.4 Fouling self-peeling antifouling coatings without biocides

The main challenges with toxic coatings include their high toxicity, limited abrasion resistance, and unstable polymer degradation rates, which can adversely affect the environment, economy, and human health. The International Maritime Organization (IMO) prohibits the use of toxic substances in conventional antifouling paints through regulations. Fouling self-peeling antifouling coatings without biocides rely on fewer sites for fouling organisms to anchor themselves. After being washed by seawater, the adhering marine organisms are released into the seawater without being killed. This process is largely determined by the surface energy and elastic modulus of the coating. Therefore this type of coating has long-lasting anti-fouling properties (**Fig. 8**).

A kind of mono- (M) and tetra- (Q) functional siloxane (MQ organosilicone resin), as a new type of polymer material with a three-dimensional structure, has molecular bonds that form a high cross-linking density. Zhou et al. (2021) produced a novel copolymer of MQ silicone resin with acrylic acid, which was doped into PDMS. Through this experiment, the authors and others prepared a self-peeling antifouling coating with a self-delamination structure and ensured excellent adhesion between the coating and the substrate. Hybrid coatings containing hyperbranched triblock copolymers also offer resistance to common microbial adhesion. Such coatings provide a multifunctional synergy of surface renewal, fouling release and antifouling through internal compound cleavage (Xie et al., 2023). Also, the use of biomimetic natural antifouling compounds is one of the very popular ways of preparing coatings today. Some studies have introduced coumarin into acrylic resins, and the coatings can inhibit microbial growth, extracellular polymer aggregation, and metabolite secretion during the early stages of fouling adhesion (Hou et al., 2024).

This kind of self-peeling antifouling paint faces the same problem as the above self-peeling antifouling paint containing biological killer agent, but it has leading significance in environmental protection (Gu et al., 2020; Selim et al., 2018, 2024; Mo et al., 2022).

5.3.5 Self-polishing or other types of antifouling coatings without biocides

The earliest traditional self-polishing coatings are mainly embedded with biological killing agents such as cuprous oxide and organotin copolymers, especially TBT, which shows high killing efficiency on fouling organisms. However, due to the extreme toxicity of TBT, international organizations have decided to completely ban its application. Currently, researches focus on environmentally friendly materials that can achieve the same antifouling strategies of organotin coatings, among which tin-free self-polishing coatings occupy an important position in the market (Wu et al., 2024a).

A functional resin with good antimicrobial properties was prepared by synthesising indole derivatives with good bioactivity and introducing them into zinc acrylate resin. The coating also has good bioinhibitory activity (Dong et al., 2024). Wang et al. (2022) were the first to create polydimethylsiloxane coatings with high transparency and excellent surface homogeneity by fabricating bionic smooth surfaces, offering a reliable strategy for the anti-fouling technology of optical windows in marine instruments. Furthermore, Zhang et al. (2017) produced three different boundary structures using a bionic replica method and continued to investigate the bionic antifouling mechanism under the advantage of siloxane-modified resin coatings, utilizing the microstructure to pave the way for future development.

As shown in **Fig. 9**, in recent years, the design of anti-fouling coatings has tended to focus more on modifying their surface properties to be biocide-free. In brief, the means of controlling biofouling have evolved in a green and sustainable direction (Ali et al., 2024; Li et al., 2024). Biocides-free coatings show significant advantages in terms of environmental friendliness, health protection, durability and wide applicability. Choosing environmentally friendly paints is not only responsible for your health, but also contributes to environmental protection.

6 Conclusions and perspective

As China accelerates its development into a maritime power, marine biofouling has become a complex and urgent issue that needs to be addressed. This is particularly true for marine equipment operating in severe marine environments with extreme conditions, where safe and

efficient anti-fouling technologies are crucial. A summary of existing anti-fouling technologies for marine environments is presented in **Fig. 9**. Building upon traditional marine antifouling techniques, it is essential to enhance the fundamental properties of materials, physical removal technologies, and environmentally friendly antifouling coatings. These improvements should enable the equipment to resist biological attachment in harsh marine conditions and simultaneously satisfy the demands for enhanced corrosion resistance, abrasion resistance, and weather resistance under such conditions.

As depicted in **Fig. 9**, in recent years, the heightened awareness of environmental protection and advancements in science and technology have spurred further innovation in biofouling control technology, building upon its original foundations. However, there remains significant potential for the development of biofouling control technologies tailored for severe marine environments. Looking ahead, it is anticipated that the materials used in marine facilities will undergo modification at the microscopic level, focusing on structural properties. Research will likely concentrate on developing coatings that are not only multifunctional but also environmentally friendly. The aim is to achieve the long-term and effective control of biofouling.

Chemical antifouling technologies containing emerging materials dominate in antifouling coatings. In addition to common nanomaterials and modified compounds, graphene, carbon nanotubes, and molecular sieves have been used. Due to its excellent electrical, thermal and mechanical properties, graphene can improve the antifouling performance and durability of coatings. Graphene has certain hydrophobicity and high stability. But the high cost of graphene technology is difficult to solve. Therefore, these emerging compounds cannot be applied on a large scale at present.

With the development of artificial intelligence, many scholars and companies have already developed big data monitoring techniques. Offshore monitoring platforms have also been developed to some extent. However, such technologies have the same problems of high cost and difficulty in popularization. And, although the degree of biofouling can be simulated and predicted. Anti-fouling control technology is still a major challenge to overcome.

Selektope has a wide range of applications in marine antifouling coatings. As an anti-barnacle active antifouling agent, Selektope is added to the coating to effectively prevent barnacle adhesion. By using Selektope, marine antifouling coatings can become low in volatile organic compounds (VOC) release, reducing the impact on the environment. Due to its biodegradable nature, Selektope poses less of a burden to the environment during disposal and use. Selektope has been integrated into new antifouling coating brands in Japan and Korea.

The strategy to improve the performance of environmentally friendly marine antifouling coatings is a comprehensive issue involving multiple fields such as materials science, chemical engineering and environmental protection. The environmental impact can be reduced to a certain extent through appropriate material selection and surface design. Low surface energy materials or nanostructures such as polydimethylsiloxane, certain fluorides can effectively reduce bio-attachment in the absence of biocides. Secondly, it is also very important to develop new environmentally friendly antifouling agents and further optimize coating formulations. For example, reducing the use of biocides and concentrating on enhancing the self-polishing properties of coatings. Finally, it is crucial to develop innovative and integrated antifouling technologies. Today a very large number of scholars have studied the application of composite materials or multifunctional antifouling measures. Future research in the field of antifouling will also be directed in these directions, leading to environmentally friendly antifouling systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary data

Supplementary data associated with this article can be found in the online version at xxxxxx.

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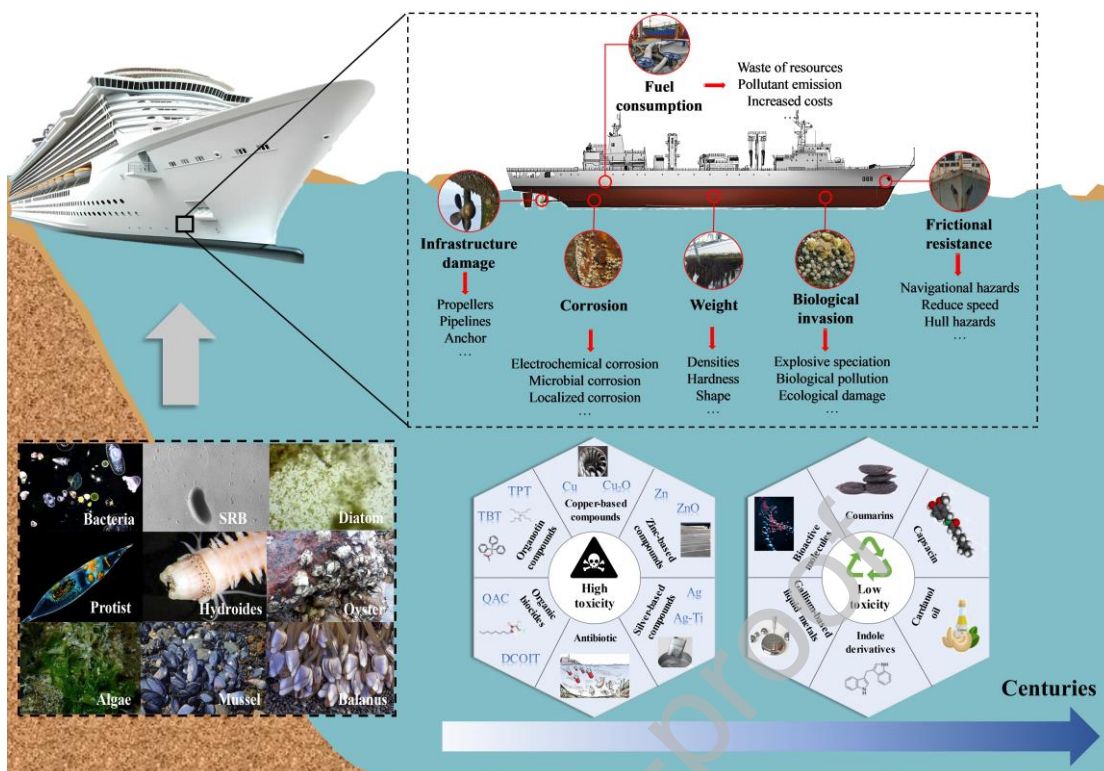


Fig. 1 Research of biological fouling in marine environment.

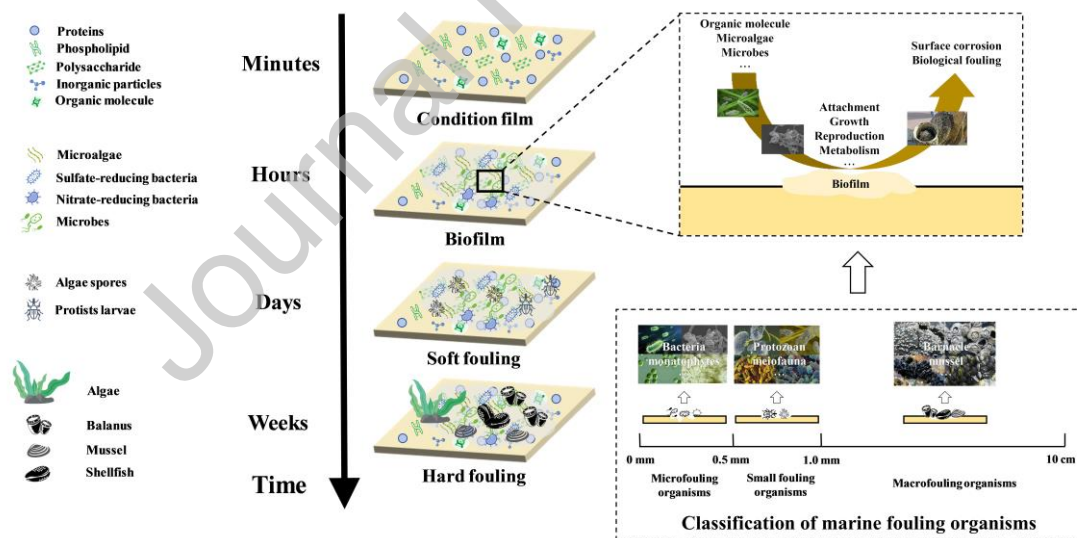


Fig. 2 The formation process of biological fouling.

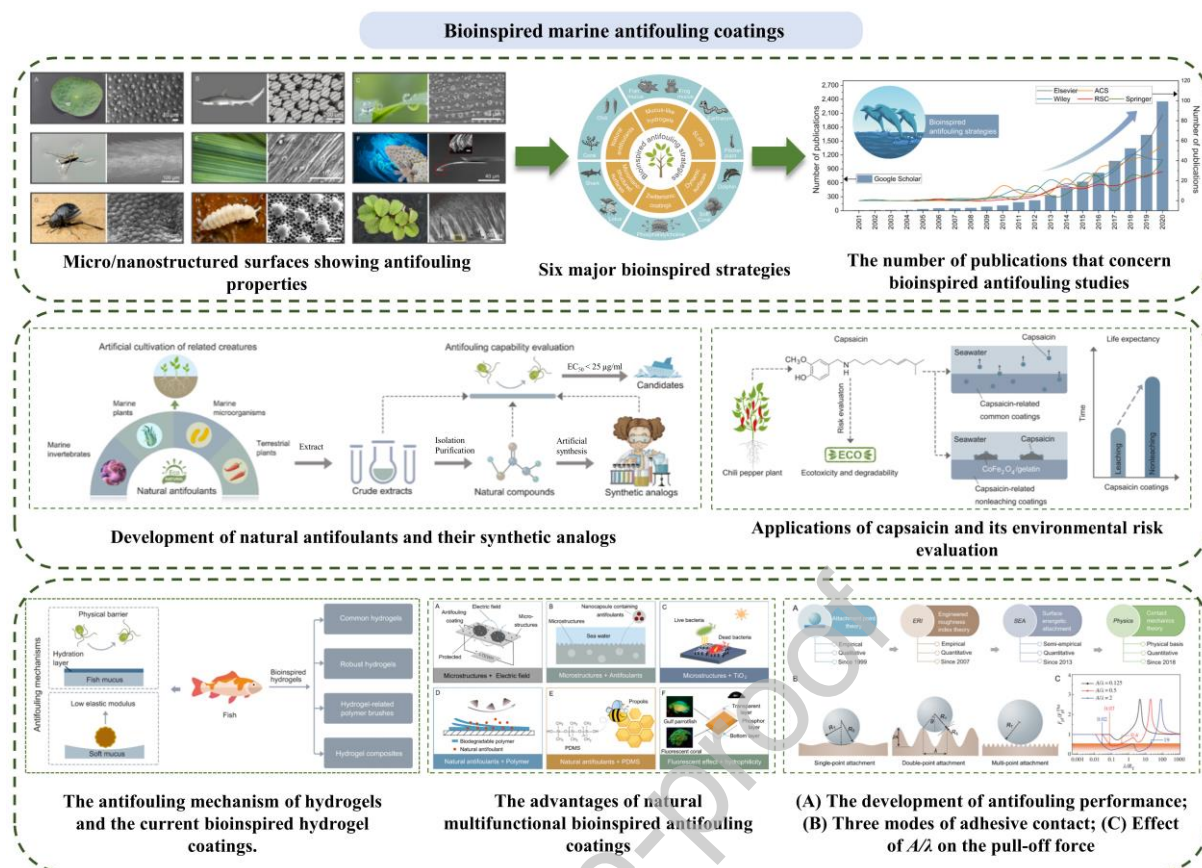


Fig. 3 Examples of bioinspired marine antifouling coatings (Jin et al., 2022).

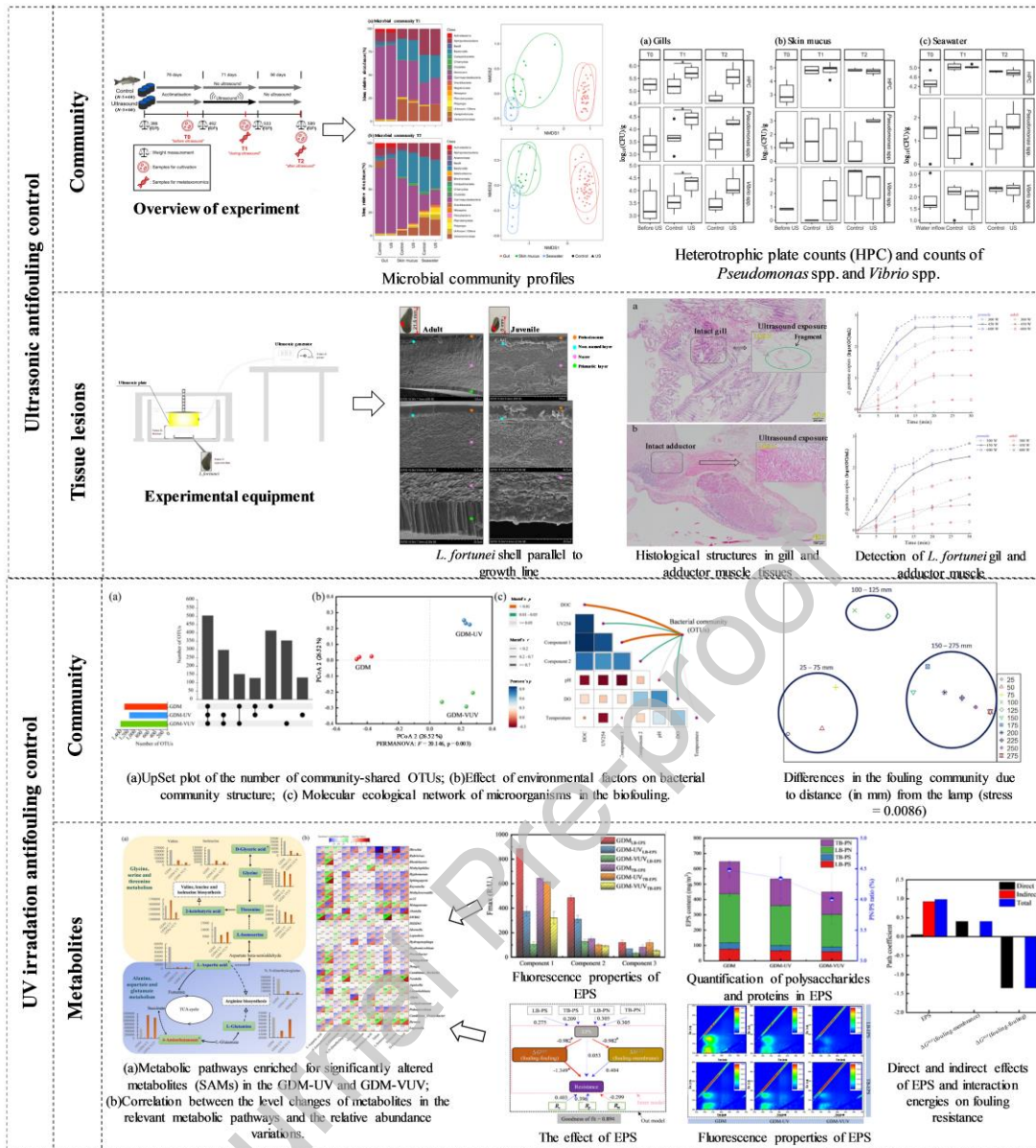


Fig. 4 Impact of physical control technology on marine biofouling (Zhou et al., 2020; Feng et al., 2023; Braga et al., 2020b).

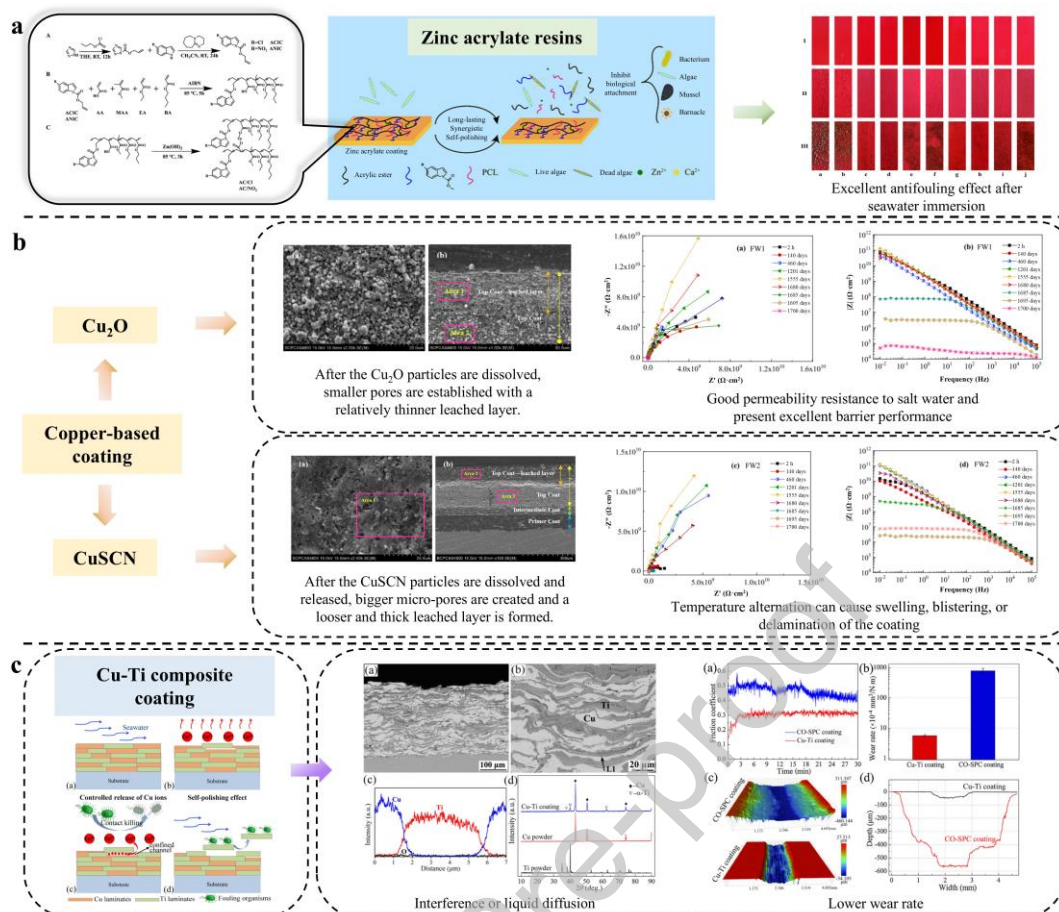


Fig. 5 Examples of self-polishing coatings with biocides: (a) novel indole-based self-polishing acrylic antifouling coatings; (b) two self-polishing antifouling coating systems with copper-based antifouling agents; (c) a kind of durable self-polishing antifouling Cu-Ti coating (Hu et al., 2024; Zhang et al., 2022; Tian et al., 2021).

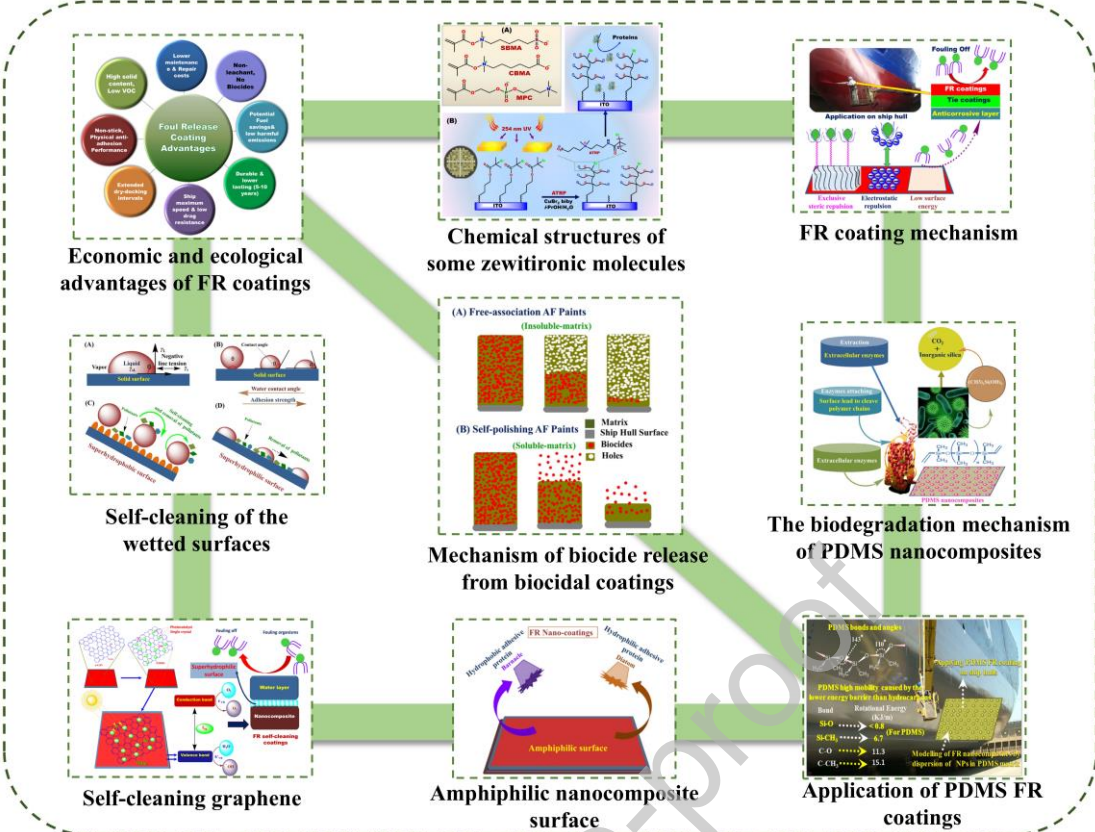


Fig. 6 Recent progress in marine polymeric nanocomposite coatings (Selim et al., 2017).

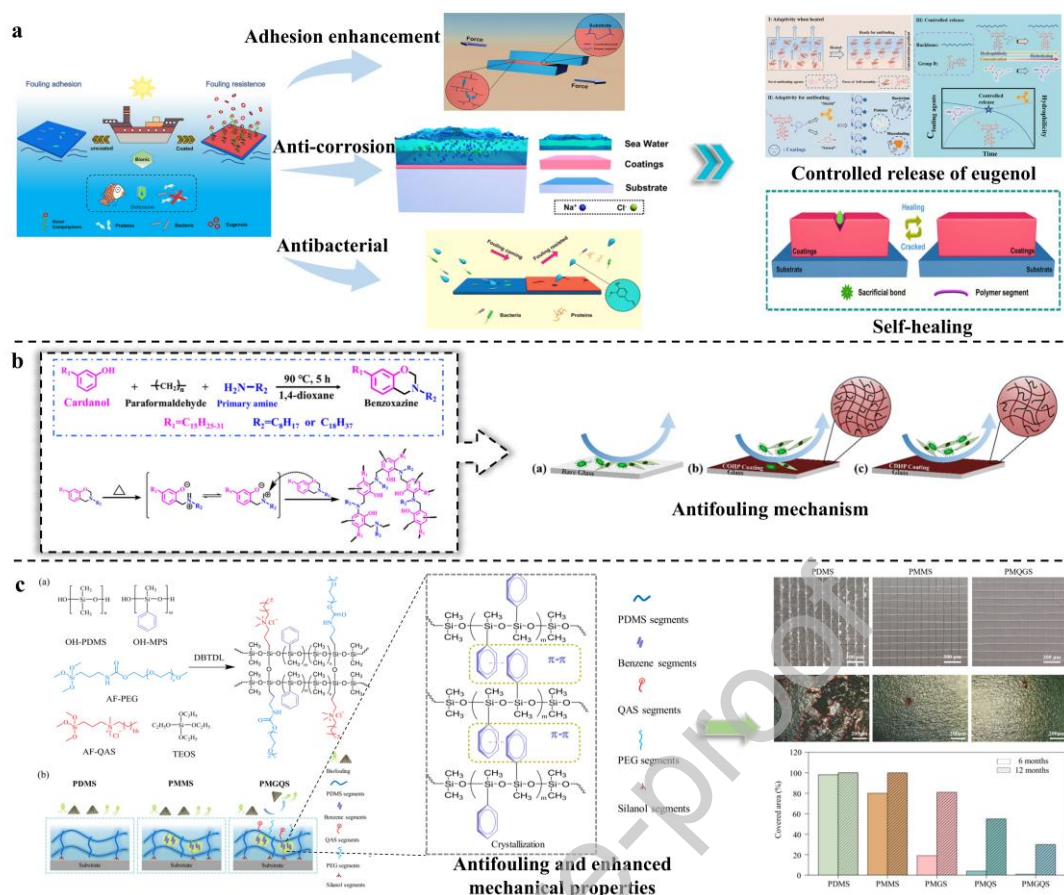


Fig. 7 Examples of self-peeling coatings with biocides: (a) a kind of high-efficiency self-healing coating featuring robust controlled release of eugenol; (b) molecular structure design of polybenzoxazines with low surface energy and low modulus; (c) a mechanically robust silicon-based cross-linking polymer for the sustainable marine antifouling coatings (Liu et al., 2024a; An et al., 2024). PDMS: polydimethylsiloxane; QAS: quaternary ammonium salts; PEG: polyethylene glycol; AF-PEG: alkoxy silane-functionalized polyethylene glycol; OH-MPS: bis-silanol-terminated methyl phenyl silicone; DBTDL: ditin butyl dilaurate; OH-PDMS: OHX-4010 silicone polymer resin; TEOS: tetraethoxysilane; PMMS: polymethylmethoxysilane.

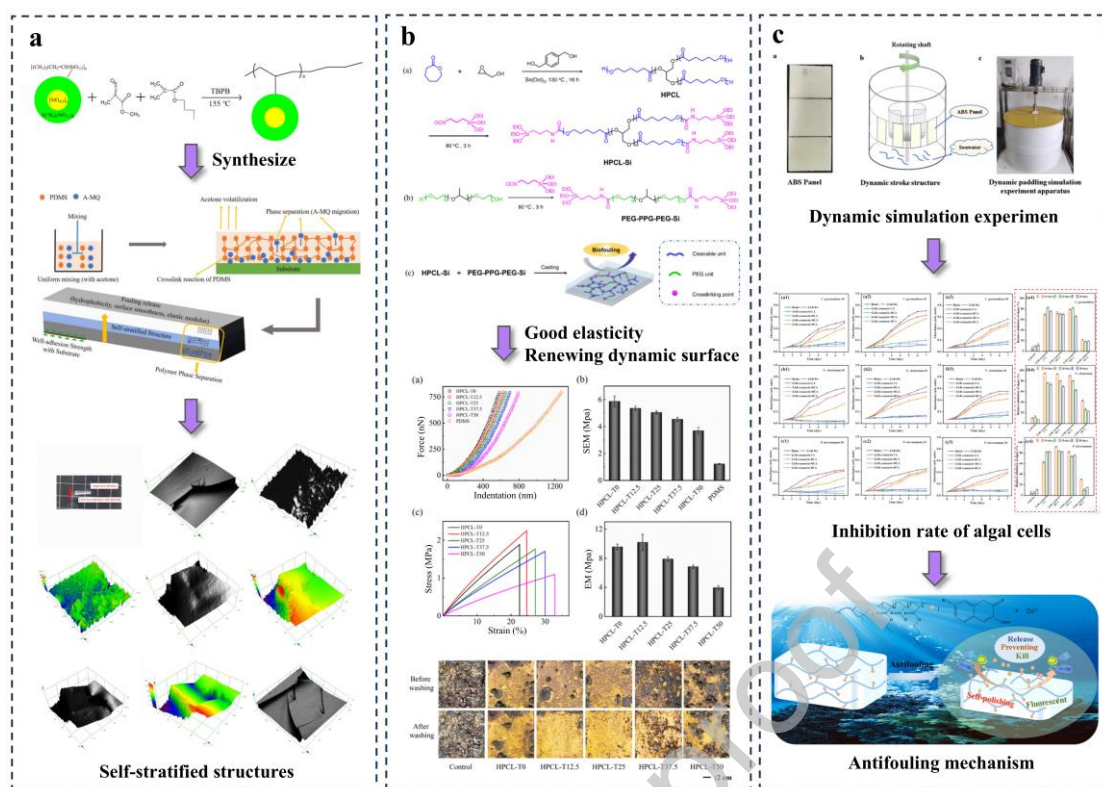


Fig. 8 Examples of self-peeling coatings without biocides: (a) self-stratified fouling release coatings based on polydimethylsiloxane incorporated with acrylate-MQ silicone copolymer; (b) non-silicone elastic coating with fouling resistance and fouling release abilities based on degradable hyperbranched polymer; (c) fluorescent zinc acrylate resins containing coumarin structures with enhanced antifouling performance (Zhou et al., 2021; Hou et al., 2024).

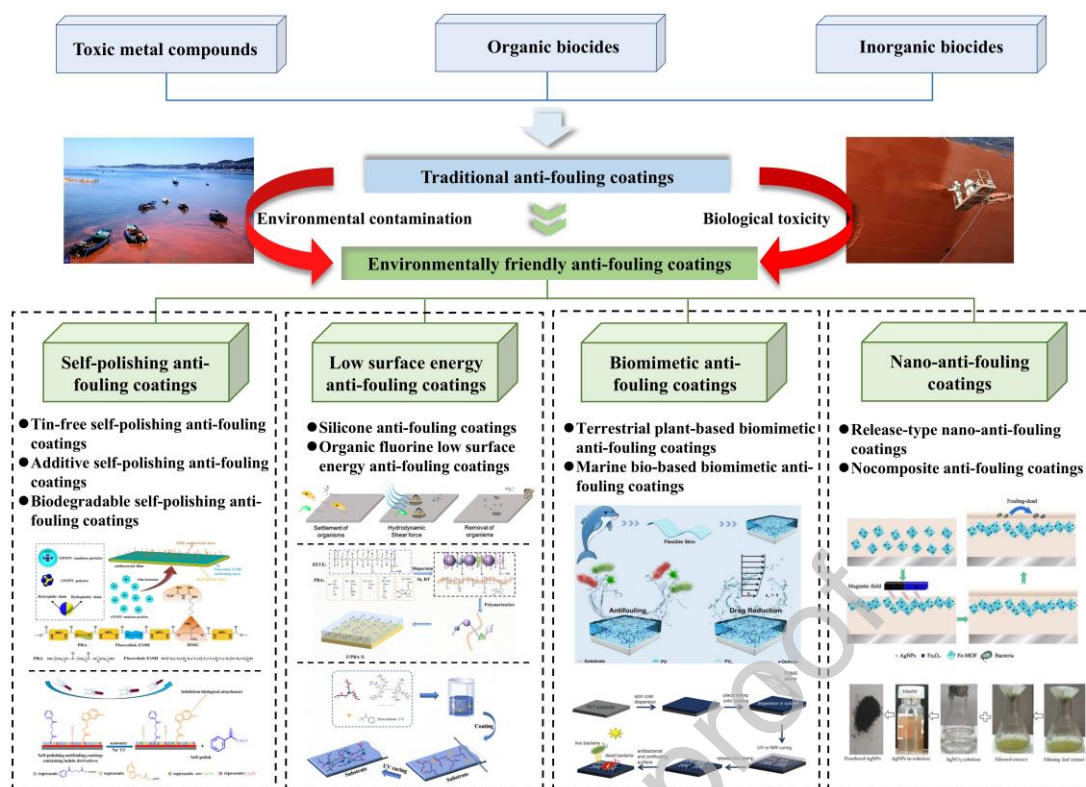
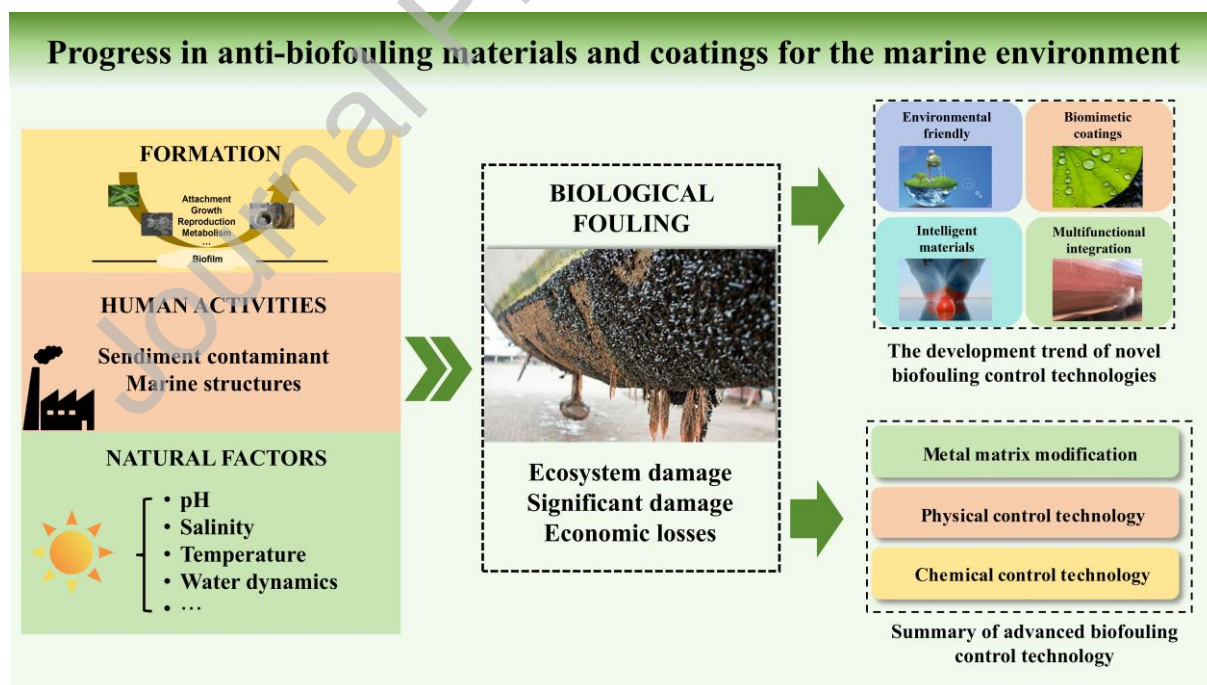


Fig. 9 Development of marine antifouling coatings (Wu et al., 2024a; Dong et al., 2024).



Graphical abstract

Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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